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Saveetha Institute of Medical And Technical Sciences
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Food Rescue Connect: Smart Food Donation and Redistribution Management System

Industry Context

Agile sprint planning & user story workshop

A CASE STUDY REPORT

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1. ABSTRACT

Food waste has become one of the most serious global challenges affecting environmental sustainability, economic development, and food security. Every year, millions of tons of edible food are discarded by restaurants, hotels, supermarkets, catering services, event organizers, and food manufacturers, while at the same time millions of people continue to suffer from hunger and malnutrition. The major reason behind this imbalance is the lack of an efficient coordination system that can connect food donors with charitable organizations and beneficiaries in real time. Existing food donation practices often depend on manual communication through phone calls, social media, or messaging applications, resulting in delayed responses, food spoilage, and unequal distribution of available food resources.

FoodRescue Connect is proposed as an intelligent Food Donation and Redistribution Management System that utilizes Artificial Intelligence (AI), Cloud Computing, Internet of Things (IoT), and GPS technologies to automate and optimize the food donation process. The platform provides a centralized digital environment where restaurants, hotels, supermarkets, NGOs, volunteers, and beneficiaries can collaborate efficiently. Food donors can register surplus food by providing details such as food type, quantity, preparation time, expiry time, and pickup location. NGOs and beneficiaries can search for nearby food donations based on availability and urgency, while volunteers receive optimized pickup and delivery routes through GPS navigation.

The proposed system continuously monitors food availability, quantity, expiry time, and transportation status. Artificial Intelligence algorithms analyze historical donation records and real-time requests to predict food demand, recommend the most suitable beneficiaries, and prioritize urgent distribution requests. Route optimization algorithms reduce transportation time and operational costs while ensuring that food reaches beneficiaries before its quality deteriorates.

The cloud-based architecture enables centralized storage of donation records, user information, volunteer activities, food distribution history, and analytical reports. Real-time dashboards provide complete visibility of ongoing donations, food collection, delivery status, volunteer assignments, and overall platform performance. Automated notifications are sent to donors, volunteers, NGOs, and beneficiaries whenever new donations are created, accepted, collected, or delivered.

FoodRescue Connect also generates analytical reports that help government agencies and charitable organizations evaluate food waste reduction, donation trends, volunteer performance, and community impact. These reports support better decision-making and future planning for food recovery initiatives.

By integrating modern digital technologies with intelligent decision-making, FoodRescue Connect significantly reduces edible food waste, improves food accessibility for vulnerable communities, enhances operational efficiency, and promotes sustainable development. The proposed platform serves as a scalable, transparent, and socially responsible solution for building an intelligent food redistribution ecosystem that benefits society while contributing to environmental conservation.

2. INTRODUCTION

Food is one of the most essential resources required for human survival and overall well-being. Despite significant advancements in agriculture, food production, and supply chain management, millions of people around the world continue to experience hunger and food insecurity. According to international food organizations, a substantial amount of perfectly edible food is wasted every day due to inefficient distribution systems, improper inventory management, overproduction, and lack of coordination between food donors and charitable organizations. Restaurants, hotels, supermarkets, event organizers, food manufacturers, and catering services frequently dispose of surplus food because there is no efficient mechanism to identify nearby organizations or individuals who can utilize the food before it expires.

Food waste has serious economic, environmental, and social consequences. When edible food is discarded into landfills, it contributes significantly to greenhouse gas emissions, particularly methane, which accelerates climate change. Food production also consumes valuable natural resources such as water, electricity, fuel, labor, and agricultural land. Wasting food means wasting all these resources as well. At the same time, millions of underprivileged families, homeless individuals, orphanages, old-age homes, and disaster-affected communities struggle to obtain nutritious meals. Therefore, reducing food waste while improving food accessibility has become a major global sustainability objective.

Traditional food donation systems are primarily based on manual communication through telephone calls, emails, messaging applications, or social media platforms. These methods often involve lengthy coordination between donors, NGOs, volunteers, and beneficiaries.

Recent technological advancements provide an opportunity to overcome these challenges. Cloud computing enables centralized storage and real-time sharing of donation information across multiple organizations. GPS technology supports efficient route planning for volunteers, reducing travel time and transportation costs. IoT-based sensors can monitor food storage conditions such as temperature and humidity to ensure food quality during transportation. Artificial Intelligence and Machine Learning algorithms can analyze historical donation data to predict food demand, estimate food availability, recommend suitable beneficiaries, and optimize resource allocation.

FoodRescue Connect has been designed as an intelligent food donation and redistribution management platform that integrates all these technologies into a single digital ecosystem. The platform connects food donors, NGOs, volunteers, government organizations, and beneficiaries through a centralized cloud-based application. It automates food donation registration, volunteer assignment, delivery tracking, quality monitoring, and analytical reporting. AI-powered algorithms optimize food allocation and transportation while minimizing operational delays and food spoilage.

The proposed system also provides complete transparency throughout the donation lifecycle. Stakeholders can monitor food availability, collection status, delivery progress, and beneficiary information using interactive dashboards. Real-time notifications ensure timely communication between all participants, while analytical reports help organizations measure their social impact, donation performance, and contribution toward reducing food waste.

Overall, FoodRescue Connect transforms traditional food donation into an intelligent, technology-driven process that supports environmental sustainability, community welfare, and efficient food resource management.

3. PROBLEM STATEMENT AND EXISTING CHALLENGES

The rapid growth of the food service industry has significantly increased the amount of edible food that is discarded every day. Restaurants, hotels, supermarkets, catering services, bakeries, food manufacturers, and event organizers frequently generate surplus food that remains safe for consumption but is ultimately disposed of due to the absence of an efficient redistribution mechanism. At the same time, millions of people including homeless individuals, orphanages, old-age homes, low-income families, and disaster-affected communities continue to experience food shortages. This imbalance between food surplus and food demand has become a major social and environmental challenge.

One of the primary problems associated with existing food donation systems is the heavy dependence on manual communication. Food donors usually contact charitable organizations through phone calls, emails, or social media platforms to inform them about available food. Similarly, NGOs spend considerable time contacting donors to locate available food donations. This manual coordination often leads to communication delays, confusion, duplicate requests, and missed donation opportunities.

Another major challenge is the absence of real-time information regarding food availability. Many organizations are unaware of nearby food donations because there is no centralized platform that continuously updates available food, donor locations, food quantity, preparation time, and expiry information. As a result, edible food often expires before it can be collected and distributed to beneficiaries.

Transportation management also creates significant operational difficulties. Volunteers frequently travel long distances without optimized routes, increasing transportation costs and delivery time. Poor route planning may result in food spoilage before delivery, especially for highly perishable food items. Existing

systems rarely utilize GPS-based intelligent route optimization to reduce travel distance and improve delivery efficiency.

Food quality monitoring is another critical concern. Most current food donation platforms provide limited information regarding food freshness, storage conditions, or expiry time. Without proper monitoring, beneficiaries may receive food that no longer meets acceptable quality standards. The absence of automated quality monitoring reduces trust among both donors and beneficiaries.

Another important limitation is the lack of Artificial Intelligence for intelligent decision-making. Existing systems generally allocate food manually without considering factors such as beneficiary demand, food quantity, transportation distance, urgency, volunteer availability, or previous donation history. Consequently, some organizations receive excess food while others receive insufficient support.

Transparency is also limited in conventional food donation systems. Donors often cannot track whether their donated food has been collected, transported, or successfully delivered to beneficiaries. NGOs similarly experience difficulty monitoring volunteer activities and overall distribution progress. Without proper tracking mechanisms, accountability and trust between stakeholders decrease significantly.

Furthermore, most existing platforms generate only basic reports that provide limited information about donation activities. Organizations require detailed analytics regarding food waste reduction, volunteer performance, beneficiary distribution, carbon emission reduction, and operational efficiency to improve future planning and policy decisions.

The major challenges of existing systems can therefore be summarized as follows:

Challenge	Existing Limitation
Manual Communication	Slow coordination between donors and NGOs
No Real-Time Updates	Delayed food collection
Poor Route Planning	Increased transportation time and cost
No AI-Based Allocation	Unequal food distribution
Limited Food Quality Monitoring	Increased risk of spoiled food
Lack of Transparency	Difficult to monitor donation lifecycle
Limited Reporting	Poor decision making
No Centralized Platform	Information scattered across multiple sources

These challenges clearly indicate the need for an intelligent food donation management platform capable of providing centralized coordination, AI-based food allocation, GPS-supported transportation planning, real-time monitoring, automated notifications, and analytical reporting.

OBJECTIVES AND SCOPE

Objectives

The primary objective of FoodRescue Connect is to develop a smart and centralized platform that enables efficient food donation and redistribution through intelligent automation and real-time coordination. The system aims to reduce food waste while ensuring that surplus food reaches deserving beneficiaries safely and quickly.

The specific objectives of the proposed system are:

Develop an intelligent cloud-based platform for managing food donation activities.

Connect restaurants, hotels, supermarkets, food manufacturers, NGOs, volunteers, and beneficiaries through a centralized application.

Monitor food availability, quantity, preparation time, expiry time, and quality in real time.

Apply Artificial Intelligence algorithms to optimize food allocation based on demand, urgency, distance, and volunteer availability.

Use GPS technology to identify nearby volunteers and generate optimized pickup and delivery routes.

Provide automated notifications for donation requests, volunteer assignments, collection status, and delivery updates.

Track the complete lifecycle of every food donation from registration to beneficiary delivery.

Generate analytical reports showing food waste reduction, donation statistics, volunteer performance, and environmental impact.

Improve transparency and accountability throughout the food donation process.

Promote sustainable food recovery practices and support community welfare using digital technologies.

Scope of the System

FoodRescue Connect covers multiple operational modules required for intelligent food redistribution.

The system includes user registration and authentication for donors, NGOs, volunteers, beneficiaries, and administrators. Authorized users can securely access platform services according to their assigned roles.

The donation management module allows donors to register available food with complete information regarding food category, quantity, preparation time, expiry time, storage condition, and pickup location

The volunteer management module assigns nearby volunteers using AI recommendations and GPS location tracking.

The AI module analyzes historical donation records, beneficiary requests, transportation distance, food expiry time, and demand patterns to optimize food allocation decisions

The IoT integration module supports food quality monitoring by collecting environmental data such as storage temperature and humidity whenever IoT sensors are available.

The dashboard provides real-time monitoring of donations, volunteer assignments, collection progress, completed deliveries, and analytical statistics.

The reporting module generates comprehensive reports that assist NGOs, government organizations, and administrators in evaluating operational performance.

System Scope Includes

User Registration and Authentication

Food Donation Management

NGO Request Management

Volunteer Assignment

AI-Based Food Allocation

GPS Route Optimization

IoT Food Quality Monitoring

Real-Time Tracking

Dashboard Visualization

Notification Management

Report Generation

Administrative Management

System Does Not Include

Physical cooking or food preparation.

Automatic food packaging.

Financial payment processing.

Warehouse inventory management.

Medical inspection of donated food.

Physical transportation services beyond volunteer coordination.

Disposal of expired food.

The scope clearly defines the operational boundaries of FoodRescue Connect while ensuring that the platform remains focused on intelligent food donation management, efficient redistribution, and sustainable community welfare.

STAKEHOLDER ANALYSIS

A successful food donation management system must satisfy the needs of multiple stakeholders who participate in the food donation and redistribution process. FoodRescue Connect has been designed to provide a centralized platform where every stakeholder can interact efficiently according to their responsibilities. Understanding stakeholder requirements helps software developers identify system functionalities and ensures that the proposed solution meets real-world operational needs.

The primary stakeholder of the system is the **Food Donor**, which includes restaurants, hotels, supermarkets, bakeries, food manufacturers, catering services, and event organizers. Food donors require a simple interface to register surplus food, specify food quantity, preparation time, expiry time, pickup location, and monitor the collection status of their donations. Donors also expect transparency regarding whether their food has been successfully delivered to beneficiaries.

Non-Governmental Organizations (NGOs) act as intermediaries between food donors and beneficiaries. NGOs need a centralized dashboard where they can view available food donations, assign volunteers, monitor food collection, and coordinate food distribution. They also require reports showing completed donations and operational performance.

Volunteers are responsible for collecting food from donors and delivering it to beneficiaries. Volunteers require optimized pickup routes, navigation support, collection schedules, and delivery status updates. GPS integration helps volunteers reduce travel time while improving transportation efficiency.

Beneficiaries include orphanages, old-age homes, homeless shelters, low-income families, disaster relief centers, and charitable institutions. Beneficiaries require timely access to nutritious food based on their current demand. They should also receive notifications whenever food is allocated to them.

The **System Administrator** manages the overall platform by maintaining user accounts, stakeholder roles, donation categories, notification settings, AI configurations, and report generation. The administrator ensures that only authorized users can access system resources and manages overall system security.

Government Authorities may monitor donation statistics, food waste reduction, and environmental impact. Analytical reports generated by the platform help policymakers evaluate community welfare programs and sustainability initiatives.

The **Artificial Intelligence Module** acts as an intelligent software component that analyzes donation history, beneficiary demand, transportation distance, volunteer availability, and food expiry information to generate optimized food allocation recommendations.

The **GPS and Notification Services** are external system components that assist in route optimization and communication. GPS services calculate optimal transportation routes, while notification services send real-time alerts through SMS, email, or mobile applications.

Stakeholder Table

Stakeholder	Primary Responsibility	System Requirement
Food Donor	Register surplus food	Easy donation registration
NGO	Coordinate food distribution	View available donations
Volunteer	Collect and deliver food	GPS navigation and task assignment
Beneficiary	Receive donated food	Real-time food allocation
Administrator	Manage system	User and platform management
Government	Monitor activities	Reports and analytics
AI Engine	Intelligent allocation	Food demand prediction
GPS Service	Route optimization	Shortest delivery path
Notification Service	Send alerts	Real-time communication

7. FUNCTIONAL REQUIREMENTS

Functional requirements describe the major services provided by FoodRescue Connect. These requirements define what the system should perform to satisfy stakeholder needs.

User Management Module

The system shall allow users to register, log in securely, and access features according to their assigned roles. Authentication ensures that only authorized users access protected system resources.

Administrators shall be able to create, modify, activate, deactivate, and delete user accounts whenever necessary.

REQUIREMENT PRIORITIZATION

Requirement prioritization is an important activity in Software Engineering that helps developers identify which features should be implemented first based on business value and stakeholder requirements. Since every software project has limited development time, budget, and resources, not all features can be developed simultaneously. Therefore, FoodRescue Connect adopts the **MoSCoW prioritization technique**, which classifies system requirements into four categories: **Must Have, Should Have, Could Have, and Won't Have**.

The Must Have requirements represent the core functionalities without which the system cannot operate successfully. These include user authentication, food donation registration, AI-based food allocation, volunteer assignment, real-time tracking, and notification generation.

Should Have requirements improve the overall usability and efficiency of the system. Although these features are important, the platform can still operate without them during the initial deployment. Examples include advanced reporting, dashboard analytics, volunteer performance monitoring, and historical trend analysis.

Could Have requirements provide additional convenience and improve user experience but are not essential for the first release. These include chatbot assistance, multilingual support, advanced graphical dashboards, and personalized recommendations.

Won't Have requirements are planned for future releases and are intentionally excluded from the current implementation. Examples include drone-based food delivery, blockchain-enabled donation verification, and robotic food transportation. Implementing the project according to this priority order ensures that the essential functionalities are completed first while allowing additional enhancements to be added in future software versions.

MoSCoW Requirement Prioritization Table

Priority	Requirement
Must Have	User Registration
Must Have	Secure Login
Must Have	Food Donation Registration
Must Have	NGO Management
Must Have	Volunteer Assignment
Must Have	AI Food Allocation
Must Have	GPS Route Optimization
Must Have	Real-Time Tracking
Must Have	Notification Management
Must Have	Report Generation
Should Have	Dashboard Analytics
Should Have	Food Quality Monitoring
Should Have	Historical Reports
Should Have	Volunteer Performance Reports
Should Have	Environmental Impact Reports
Could Have	Chatbot Support
Could Have	Voice Commands
Could Have	Multi-Language Interface
Could Have	AI Recommendation Dashboard
Won't Have	Drone Delivery
Won't Have	Blockchain Verification

KANO MODEL ANALYSIS

The Kano Model is another requirement prioritization technique used to classify software features based on customer satisfaction. It divides system requirements into five categories:

- Basic Requirements
- Performance Requirements
- Excitement Requirements
- Indifferent Requirements
- Reverse Requirements

For FoodRescue Connect, the basic requirements include secure login, food donation registration, volunteer assignment, and beneficiary management. These are mandatory features expected by every stakeholder.

Performance requirements directly influence user satisfaction. Faster food allocation, optimized transportation routes, accurate AI predictions, and real-time tracking significantly improve system effectiveness.

Excitement requirements are innovative features that delight users even though they may not expect them. Examples include AI-powered food demand prediction, smart volunteer recommendations, multilingual support, and personalized dashboards.

Indifferent requirements are features that have little impact on user satisfaction, such as dashboard themes or interface color customization.

Reverse requirements are features that may reduce usability if unnecessarily complicated. Excessive notifications or overly complex dashboards may negatively affect user experience.

Kano Model Table

Category	Features
Basic	Login, Registration, Donation Management
Performance	AI Allocation, GPS Tracking, Notifications
Excitement	AI Prediction, Smart Dashboard, Analytics
Indifferent	Theme Customization
Reverse	Excessive Notifications

ACTORS AND USE CASES

An actor represents any user or external system that interacts with FoodRescue Connect. Each actor performs specific operations according to their assigned responsibilities.

The primary actor is the Food Donor, who registers surplus food and monitors donation status.

The NGO searches available donations, accepts requests, assigns volunteers, and monitors food distribution.

The Volunteer receives pickup requests, collects food, follows GPS navigation, and delivers food to beneficiaries.

The Beneficiary receives food allocations and confirms successful delivery.

The Administrator manages users, food categories, AI settings, reports, and overall platform operations.

External technical actors include the AI Engine, GPS Service, Cloud Database, and Notification Service.

System Architecture

FoodRescue Connect follows a **layered architecture** that integrates cloud computing, Artificial Intelligence (AI), GPS technology, IoT, and centralized data management to automate food donation and redistribution. The layered architecture separates the system into independent functional components, making the software scalable, maintainable, secure, and easy to upgrade. Each layer performs a specific task while communicating with the adjacent layers through well-defined interfaces. The architecture begins with the **User Layer**, which consists of food donors, NGOs, volunteers, beneficiaries, administrators, and government authorities. These stakeholders interact with the system through a web or mobile application. Users can register, log in, donate food, search available donations, monitor delivery status, generate reports, and manage system settings according to their assigned roles.

The second layer is the **Application Layer**, which processes all user requests. This layer contains modules such as user authentication, food donation management, NGO management, volunteer management, AI-based food allocation, GPS route optimization, dashboard visualization, notification management, and report generation. Every user request passes through this layer before being stored or processed.

The third layer is the **Artificial Intelligence Layer**. This layer analyzes historical donation data, beneficiary requests, volunteer availability, food expiry information, and transportation distance. AI algorithms predict food demand, recommend the most suitable beneficiaries, optimize food allocation, and reduce food wastage by prioritizing urgent donations.

The fourth layer is the **Cloud Database Layer**, where all system information is stored securely. The database contains user profiles, food donation records, volunteer assignments, beneficiary details, delivery history, AI predictions,

notification logs, and analytical reports. Cloud storage enables centralized data access and supports real-time synchronization across multiple devices.

The fifth layer consists of **External Services**, including GPS services, IoT sensors, email services, SMS gateways, and push notification systems. GPS services provide optimized transportation routes, while IoT sensors monitor food storage conditions such as temperature and humidity. Notification services ensure that stakeholders receive real-time updates regarding donation activities.

The architecture allows each component to operate independently while supporting seamless communication between all stakeholders involved in food redistribution.

AI AND IoT FOOD REDISTRIBUTION WORKFLOW

Food Rescue Connect combines Artificial Intelligence, IoT, GPS, and cloud computing to automate food redistribution.

The workflow begins when a food donor registers surplus food. The donor enters details including food type, quantity, expiry time, storage condition, and pickup location.

The donation information is immediately stored in the cloud database. The AI engine retrieves historical donation records, beneficiary requests, volunteer availability, and transportation distance.

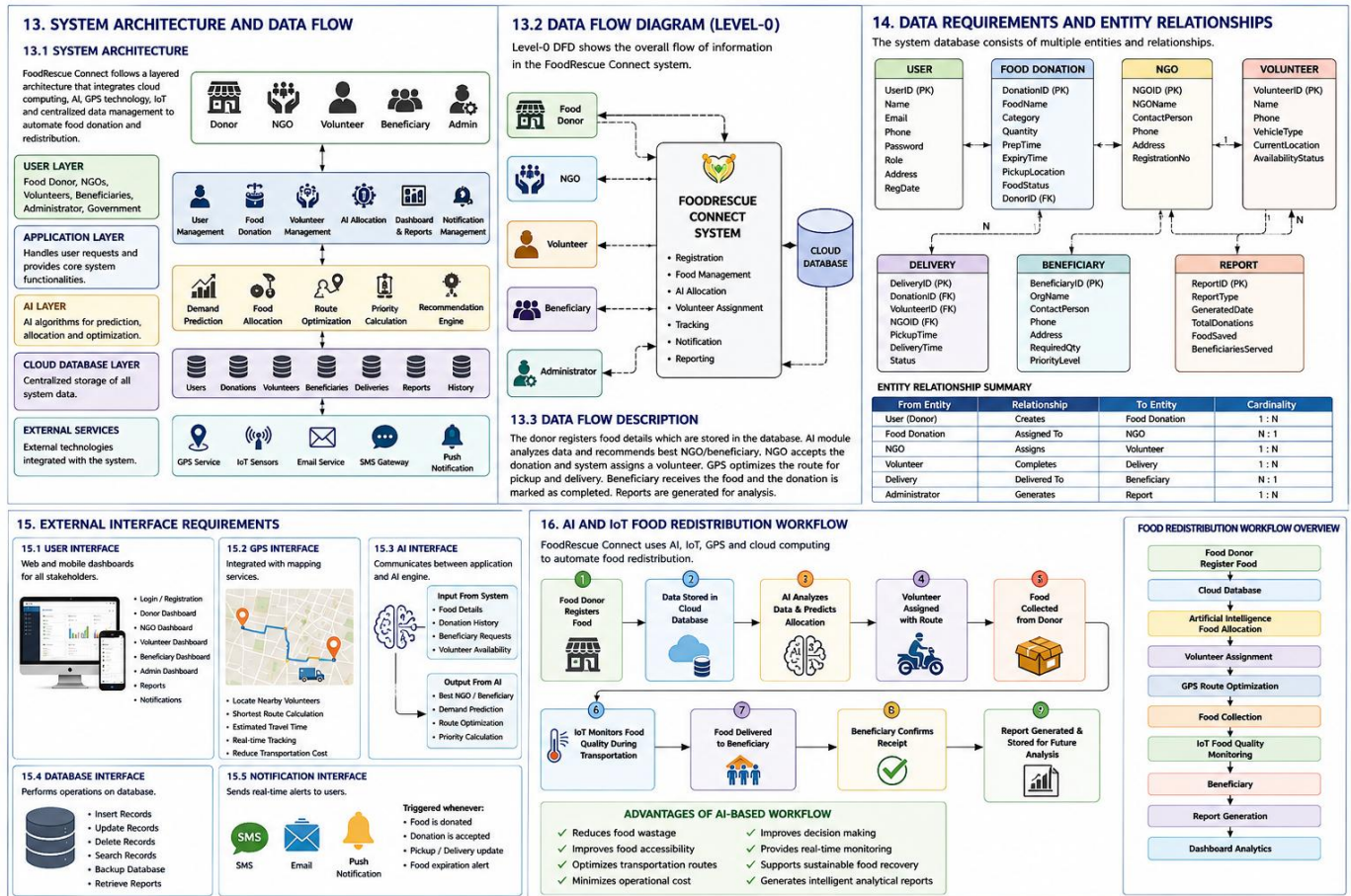
The Artificial Intelligence module performs the following activities:

- Predicts food demand.
- Calculates food priority.
- Selects nearby NGOs.
- Optimizes food allocation.
- Minimizes food wastage.

If IoT sensors are installed, they continuously monitor food temperature and storage conditions during transportation. Sensor readings are transmitted to the cloud platform where abnormal conditions generate alerts.

The volunteer receives an assignment together with GPS navigation. The shortest route is calculated automatically to reduce transportation time.

After successful delivery, the beneficiary confirms food receipt. The donation history is updated, reports are generated, and the AI model stores new data for future prediction.



JIRA PROJECT AND REQUIREMENT ORGANIZATION

Jira is one of the most widely used Agile project management tools for planning, tracking, and managing software development projects. In FoodRescue Connect, Jira is used to organize project requirements into Epics, User Stories, Tasks, and Subtasks. This hierarchical organization helps the development team monitor project progress, assign responsibilities, prioritize requirements, and ensure timely delivery of software modules.

The FoodRescue Connect project is created in Jira using the project name **"FoodRescue Connect – Smart Food Donation and Redistribution Management System"** with a project key such as **FRC**. The project contains all software requirements, sprint planning activities, bug reports, testing tasks, and deployment activities.

The overall Jira hierarchy is:

Project → Epics → User Stories → Tasks → Subtasks

This structure enables better project planning and improves communication among developers, testers, project managers, and stakeholders.

REQUIREMENT TRACEABILITY MATRIX

Requirement Traceability Matrix (RTM) ensures that every requirement is implemented, tested, and validated throughout software development.

The traceability process is:

Problem Statement → Objective → Requirement → Epic → User Story → Testing

This helps developers verify that every stakeholder requirement has been addressed.

Jira Epics

The major Epics for FoodRescue Connect are listed below.

Epic ID	Epic Name	Description
EP-01	User Management	User Registration and Login
EP-02	Food Donation Management	Manage Food Donations
EP-03	NGO Management	NGO Operations
EP-04	Volunteer Management	Volunteer Assignment
EP-05	AI Food Allocation	AI Recommendation Engine
EP-06	GPS Route Optimization	Navigation and Tracking
EP-07	Notification Management	SMS, Email & Alerts
EP-08	Dashboard & Analytics	Reports and Visualization
EP-09	System Administration	User Roles & Security

17. JIRA PROJECT AND REQUIREMENT ORGANIZATION

17.1 INTRODUCTION

Jira is used to plan, track and manage all the requirements of FoodRescue Connect project. The project is organized into Epics, User Stories, Tasks and Subtasks. This helps the team to manage work efficiently and deliver the system in a timely manner.

Jira
Project Key: FRC
Project Name: FoodRescue Connect – Smart Food Donation and Redistribution Management System

17.2 PROJECT ORGANIZATION

Project Hierarchy

```

PROJECT
├── EPICS
│   └── Major Functional Modules
├── USER STORIES
│   └── Requirements from User Perspective
├── TASKS
│   └── Development Activities
└── SUB TASKS
    └── Detailed Work Items
    
```

17.3 JIRA EPICS

Epic ID	Epic Name	Description
EP-01	User Management	User Registration, Login, Roles
EP-02	Food Donation Management	Manage Food Donations
EP-03	NGO Management	NGO Operations and Requests
EP-04	Volunteer Management	Volunteer Assignment and Tracking
EP-05	AI Food Allocation	AI Recommendation Engine
EP-06	GPS Route Optimization	Route Planning and Navigation
EP-07	Notification Management	SMS, Email and Push Notifications
EP-08	Dashboard & Analytics	Reports and Visualization
EP-09	System Administration	User Roles, Security, Settings

17.4 EPIC DESCRIPTION

- EP-01 User Management**
Includes user registration, login, authentication, profile management and role based access.
- EP-02 Food Donation Management**
Allows donors to add, update, view and track food donations.
- EP-03 NGO Management**
NGOs can search donations, accept requests and manage beneficiaries.
- EP-04 Volunteer Management**
Handles volunteer registration, assignment, tracking and performance.
- EP-05 AI Food Allocation**
AI models predict food demand and recommend best beneficiaries.
- EP-06 GPS Route Optimization**
Integrates maps to provide shortest routes and track deliveries.
- EP-07 Notification Management**
Sends alerts through SMS, Email and Push Notifications.
- EP-08 Dashboard & Analytics**
Provides dashboards, charts and analytical reports.
- EP-09 System Administration**
Admin manages users, roles, settings, backup and audit logs.

18. USER STORIES AND ACCEPTANCE CRITERIA

Story ID	User Story	Acceptance Criteria	Priority
US-01	As a new user, I want to register on the platform so that I can access the system.	- Registration form validates all fields. - Email must be unique. - Password is encrypted. - Confirmation message is shown.	High
US-02	As a registered user, I want to login to the platform so that I can access my dashboard.	- Correct credentials allow login. - Invalid credentials show error. - Role based dashboard is displayed.	High
US-03	As a food donor, I want to register surplus food so that NGOs can collect it.	- Food name is required. - Quantity, expiry time and location are required. - Donation is stored successfully.	High
US-04	As an NGO, I want to accept available donations so that food reaches beneficiaries.	- Donation must be available. - NGO must be authenticated. - Volunteer is assigned.	High
US-05	As a volunteer, I want GPS navigation so that I can pickup food quickly.	- Shortest route is displayed. - Pickup is confirmed. - Delivery is started.	Medium
US-06	As a volunteer, I want to update delivery status so that all can track the food.	- Delivery is completed. - Beneficiary confirms. - Status is updated.	High
US-07	As the AI Engine, I want to analyze donation information so that food reaches right beneficiaries.	- Demand is predicted. - Distance is calculated. - Priority is assigned. - Recommendation generated.	High
US-08	As an administrator, I want analytical reports so that platform performance can be evaluated.	- Report is generated. - Export (PDF/Excel) supported. - Charts are displayed.	Medium

USER STORY SUMMARY

Priority	Count	Percentage
High	6	75%
Medium	2	25%
Low	0	0%

19. REQUIREMENT TRACEABILITY MATRIX

RTM ensures that every requirement is implemented, tested and validated.
Requirement → Epic → User Story → Testing

Requirement ID	Requirement Description	Epic ID	User Story ID	Priority	Validation / Test Case
FR-01	User Registration	EP-01	US-01	High	TC-01 Registration Test
FR-02	User Login	EP-01	US-02	High	TC-02 Login Test
FR-03	User Roles & Access	EP-01	US-02	High	TC-03 Role Access Test
FR-04	Food Donation Registration	EP-02	US-03	High	TC-04 Donation Test
FR-05	Update/Delete Donation	EP-02	US-03	Medium	TC-05 Update Test
FR-06	Search Donations	EP-02	US-03	High	TC-06 Search Test
FR-07	Accept Donation	EP-03	US-04	High	TC-07 Accept Test
FR-08	Volunteer Assignment	EP-04	US-05	Medium	TC-08 Assignment Test
FR-09	GPS Route Optimization	EP-06	US-05	Medium	TC-09 GPS Test
FR-10	Pickup Confirmation	EP-04	US-05	Medium	TC-10 Pickup Test
FR-11	Delivery Confirmation	EP-04	US-06	High	TC-11 Delivery Test
FR-12	AI Food Allocation	EP-05	US-07	High	TC-12 AI Prediction Test
FR-13	Demand Prediction	EP-05	US-07	High	TC-13 Prediction Test
FR-14	Notifications	EP-07	System	High	TC-14 Notification Test
FR-15	Dashboard	EP-08	US-08	Medium	TC-15 Dashboard Test
FR-16	Reports Generation	EP-08	US-08	Medium	TC-16 Report Test
FR-17	User Management	EP-09	Admin	High	TC-17 User Mgmt Test
FR-18	Backup & Security	EP-09	Admin	High	TC-18 Security Test

20. JIRA WORKFLOW AND REQUIREMENT TRACKING

20.1 JIRA WORKFLOW

```

TO DO → IN PROGRESS → CODE REVIEW → TESTING → DONE
Requirements are identified and planned. Development team starts implementation. Code is reviewed by senior developers. QA team tests functionality and performance. Requirement is completed and approved.
    
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20.2 SPRINT PLANNING EXAMPLE

Sprint	Duration	Goals / Features
Sprint 1	2 Weeks	Login & Registration
Sprint 2	2 Weeks	Food Donation Management
Sprint 3	2 Weeks	NGO & Volunteer Management
Sprint 4	2 Weeks	AI Food Allocation
Sprint 5	2 Weeks	GPS Tracking & Notifications
Sprint 6	2 Weeks	Dashboard & Reports

20.3 BENEFITS OF JIRA

- ✓ Better project planning and tracking
- ✓ Sprint management and scheduling
- ✓ Requirement management
- ✓ Team collaboration
- ✓ Progress monitoring
- ✓ Bug tracking and resolution
- ✓ Task assignment and tracking
- ✓ Report generation
- ✓ Agile methodology support
- ✓ Faster and quality software delivery

CONCLUSION

FoodRescue Connect is an intelligent food donation and redistribution management system developed to address the growing challenges of food waste and food insecurity. The system integrates cloud computing, Artificial Intelligence (AI), GPS technology, and IoT to create a centralized platform that efficiently connects food donors, NGOs, volunteers, and beneficiaries. By digitizing the entire food donation process, the platform minimizes manual coordination, reduces delays, and ensures that surplus food reaches people in need before it expires.

The proposed system provides several important features, including real-time food availability monitoring, AI-based food allocation, route optimization, volunteer assignment, delivery tracking, and automated notifications. These features improve communication among stakeholders, reduce transportation time, and increase the efficiency of food collection and distribution. The use of predictive analytics enables the system to estimate food demand and prioritize urgent donation requests, ensuring that food resources are utilized effectively while minimizing wastage.

During the software engineering analysis, the stakeholder requirements were transformed into functional and non-functional requirements, use cases, user stories, system architecture, and Jira project management artifacts. The requirement prioritization using the MoSCoW model and the traceability matrix demonstrate a structured approach to software development, ensuring that every stakeholder requirement is properly implemented and validated. The modular architecture also makes the system scalable, secure, and easy to maintain for future enhancements.

Overall, FoodRescue Connect contributes significantly to environmental sustainability, social welfare, and efficient resource management. By reducing edible food waste, improving food accessibility for vulnerable communities, and supporting transparent donation tracking, the system promotes sustainable development and responsible food recovery practices. Future enhancements such as

blockchain-based donation verification, smart IoT food quality sensors, multilingual mobile applications, and advanced AI prediction models can further improve the platform's effectiveness. Therefore, FoodRescue Connect serves as a scalable, intelligent, and socially impactful solution for modern food donation and redistribution management.